AMRITA VISHWA VIDYAPEETHAM

SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Multi-Modal Sensor Fusion for Smart Home HAR

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Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23101	Adhikkesh S K	**Ms. Nalinadevi** Dept. of CSE	**Type 2** Implementation	**.../ 5**
CB.SC.U4CSE23102	Adithya Ajay			
CB.SC.U4CSE23105	Amrith B			
CB.SC.U4CSE23119	Gowreesh B			
CB.SC.U4CSE23236	Rishiikesh S K			

1 Problem Definition & Relevance

Problem Statement: Smart Home Human Activity Recognition (HAR) is currently limited by a privacy-accuracy trade-off, relying on intrusive monitoring or impractical wearables, while failing to generalize across diverse environments. Furthermore, existing models struggle to fuse heterogeneous data (ambient, inertial, RF). This project addresses these challenges by developing a Multi-Modal Sensor Fusion framework to ensure robust, privacy-preserving performance under domain shifts.

Utility Claim: *(Practical Application)*

- **Privacy-First Solution:** By replacing cameras with WiFi CSI and Ambient sensors, we maintain user privacy while achieving high-fidelity recognition.
- **Multi-Resident Disambiguation:** Integrating minimal wearables resolves the ”identity ambiguity” (who is doing what) that purely ambient systems suffer from.

2 Literature & Market Survey

Literature Survey (Top 10 Datasets): The following table summarizes key datasets used to benchmark HAR, categorized by sensor modality.

Type	Dataset	Description & Sensors	Critical Limitation
Ambient	**CASAS** (Cook et al.)	The gold standard for smart homes. Uses PIR motion, door, and temp sensors in real apartments.	**Identity Ambiguity:** Cannot distinguish *who* triggered a sensor in multi-resident homes.
Ambient	**ARAS** (Alemdar et al.)	Collected in 2 real houses with multiple residents using 20 ambient sensors.	**Sensor Noise:** High false positives from pets/visitors; lacks ground truth for overlapping activities.
Wearable	**UCI-HAR** (Anguita et al.)	Smartphone (Waist) accelerometer & gyroscope data for 6 basic ADLs.	**Location Blind:** Detects "Walking" but not "Where" (e.g., Kitchen vs. Bedroom).
Wearable	**Opportunity** (Roggen et al.)	72 sensors on body/objects capturing complex gestures (opening doors, drinking).	**Impracticality:** Users must wear a full body suit of sensors; unrealistic for daily living.
Wearable	**UniMiB-SHAR** (Micucci et al.)	Smartphone accel data focused on fall detection and basic activities.	**Limited Granularity:** Good for falls, but poor at distinguishing fine-grained ADLs (e.g., eating).
WiFi	**Widar 3.0** (Zheng et al.)	Large-scale WiFi CSI dataset for gesture recognition using commodity routers.	**Single User Focus:** Accuracy drops significantly when multiple users move simultaneously.
WiFi	**CSI-Bench** (Wang et al.)	Robustness benchmark across different domains (rooms/environments).	**Domain Shift:** Models trained on Room A fail in Room B without extensive retraining.
WiFi	**WIMANS** (Gu et al.)	Multi-user dataset using dual-band WiFi CSI and synchronized cameras.	**Computational Load:** Requires heavy processing to separate mixed signals from multiple bodies.
WiFi	**SignFi** (Ma et al.)	Fine-grained finger gesture recognition (Sign Language) using WiFi CSI.	**Short Range:** Only effective at very close range ($<$ 1m); not suitable for whole-home tracking.
WiFi	**TED-Net** (Wang et al.)	Focuses on privacy-preserving Fall Detection using WiFi skeleton formation.	**Hardware Specific:** Relies on specific router configurations that are hard to deploy universally.

Market Survey (Competitor Analysis): We analyzed existing commercial solutions to identify gaps in current technology.

Competitor	Primary Focus	The Research Gap
Smart Home Hubs (Google Nest, Alexa)	**Voice Command & Automation:** Excellent at "if this, then that" triggers (e.g., motion → lights on).	**Context Unaware:** They detect motion but do not understand *activity*. They cannot distinguish "Fall" from "Lying Down".
Wearable Tech (Fitbit, Apple Watch)	**Biometrics & Fitness:** Tracking heart rate, steps, and GPS outdoor location.	**Indoor Blindness:** Cannot interact with the home environment (e.g., doesn't know if you are at the stove or the sink).
Security Systems (ADT, Ring)	**Perimeter Safety:** Intrusion detection via door sensors and cameras.	**Privacy & Granularity:** Relies on invasive cameras. Non-camera sensors are binary and miss daily habits.
Elderly Monitors (LifeAlert)	**Emergency Response:** Reactive systems (button press) for falls.	**Reactive Only:** Triggers *after* an accident. Cannot predict varying behavior patterns or prevent incidents.
WiFi Sensing (Origin Wireless)	**Device-Free Sensing:** Using WiFi waves to detect presence/breathing.	**Identity Confusion:** Cannot robustly track multiple people. Signals mix, making it impossible to say *who* is where.

Table 2: Market Survey highlighting the need for Multi-Modal Fusion.

3 Proposed System Architecture

Core Solution: We propose a system integrating **ESP32 Ambient Sensors** (for location context), **WiFi CSI** (for device-free activity recognition), and **Minimal Wearables** (for identity tagging).

1. **Hybrid Deep Learning Model:** Implement a dual-branch neural network using CNNs to extract spatial features and LSTMs to capture temporal dependencies from the heterogeneous sensor data.

2. **Attention-Enhanced Fusion:** Specific Attention Mechanism within the fusion module to dynamically weight the importance of different modalities (e.g., prioritizing wearables over WiFi for fine motor tasks) before final classification.

3. **Synchronized Preprocessing Pipeline:** Establish a rigid preprocessing workflow focused on Temporal Alignment and sliding-window segmentation to standardize the varying sampling rates between ambient, wearable, and WiFi signals.

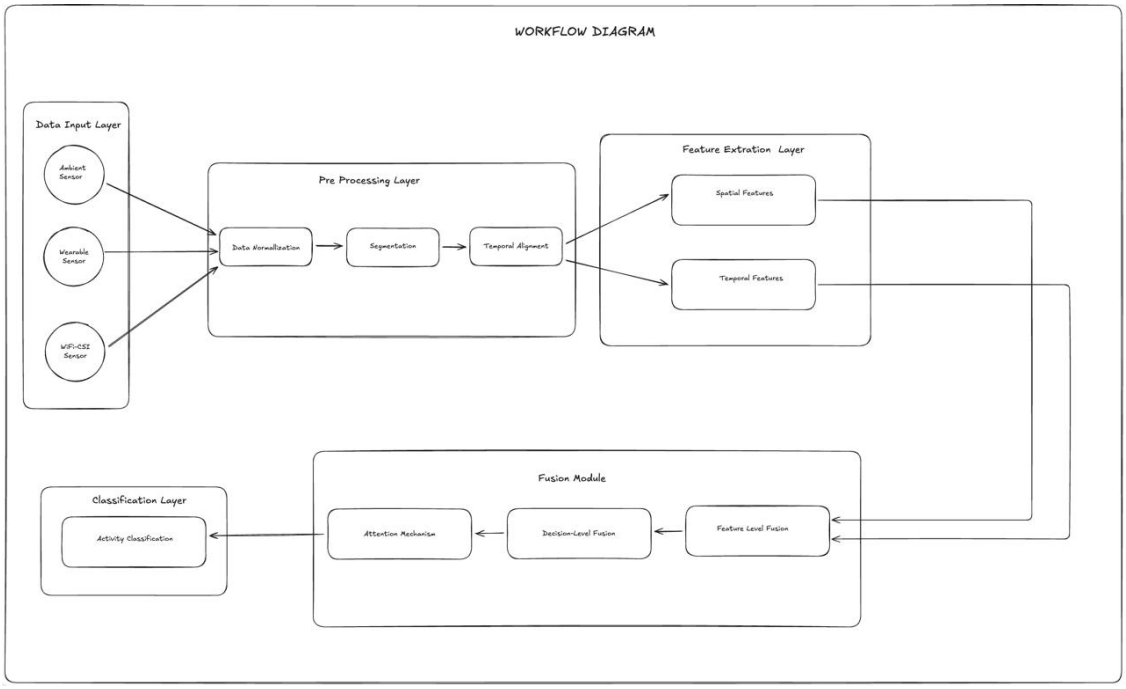


Figure 1: Proposed Multi-Modal Architecture

4 Current Status & Plan

Status: Literature survey completed; Problem formalized; Algorithm (CNN-LSTM) selected.
Next Steps: Data collection (Phase 2); Model implementation; Benchmarking.